

Task 3

Make a login in the website <http://testasp.vulnweb.com/> and intercept the network traffic using Wireshark and find the credentials that were transferred through the network.

Attack name : Network Traffic Analysis (Packet Sniffing)

Severity: High (CVSS Score: 7.5)

Reason:

Capturing network traffic on an unencrypted HTTP connection can expose sensitive information such as usernames and passwords. If login credentials are transmitted in plaintext, an attacker monitoring the network can intercept and misuse them, leading to unauthorized access.

Steps to Reproduce:

Step 1: Launch Wireshark

Open Wireshark and select the active network interface connected to the internet (eth0). Start packet capturing before accessing the target website.

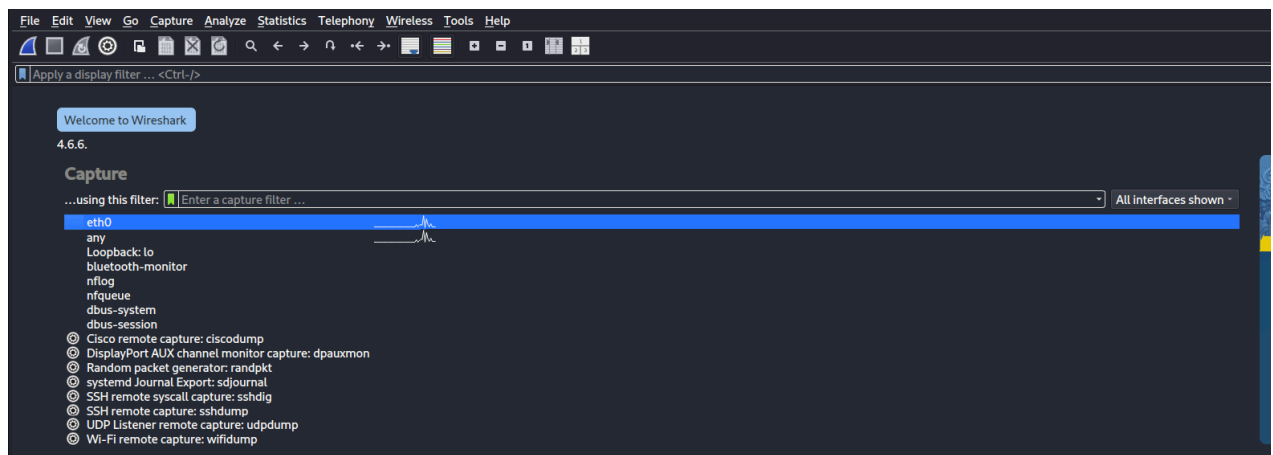


Fig 3: Wireshark Launch Page

Step 2: Login to the target website

Open the target website in a web browser and navigate to the login page.

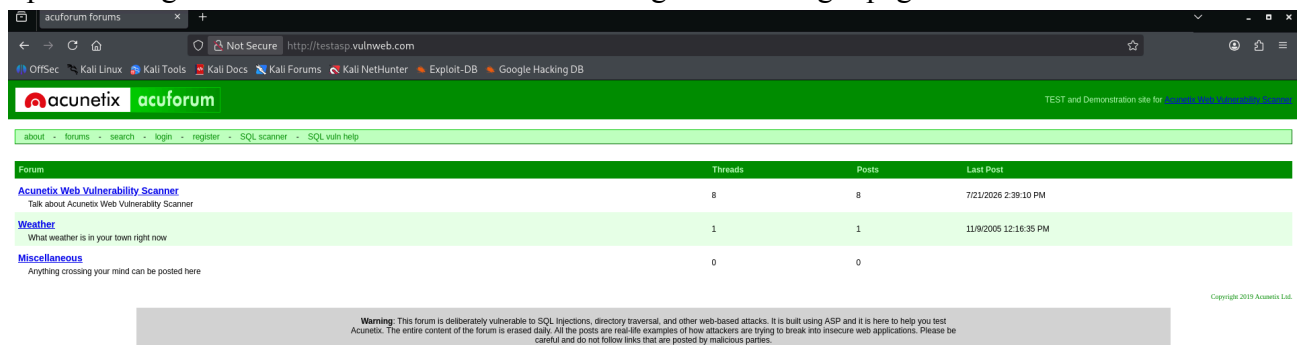


Fig 4: Target Website Homepage

Enter username and password and click Login while Wireshark continues capturing the network traffic.

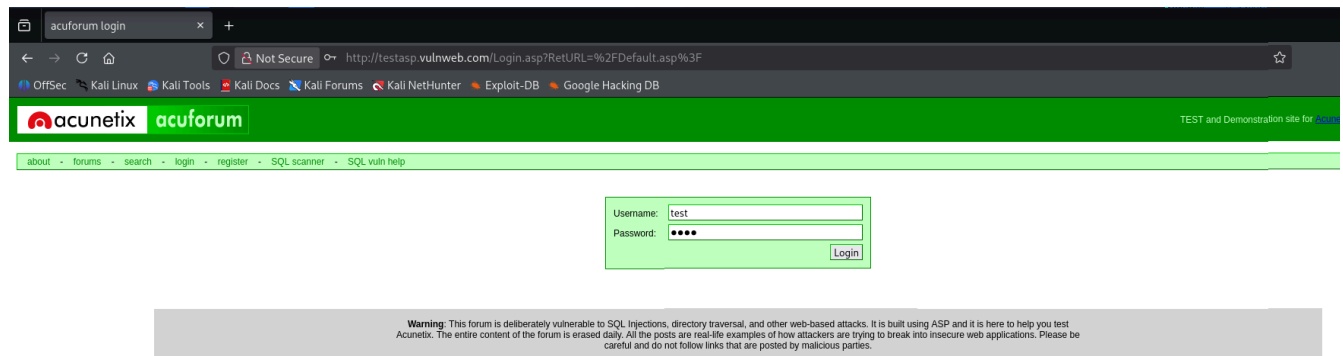


Fig 5: Login Page of the Target Website

Entered credentials:

Username: test

Password: 1234

Step 3: Capturing packets using Wireshark

The captured network packets generated during the login process are displayed in Wireshark. These packets contain the communication exchanged between the client and the target web server.

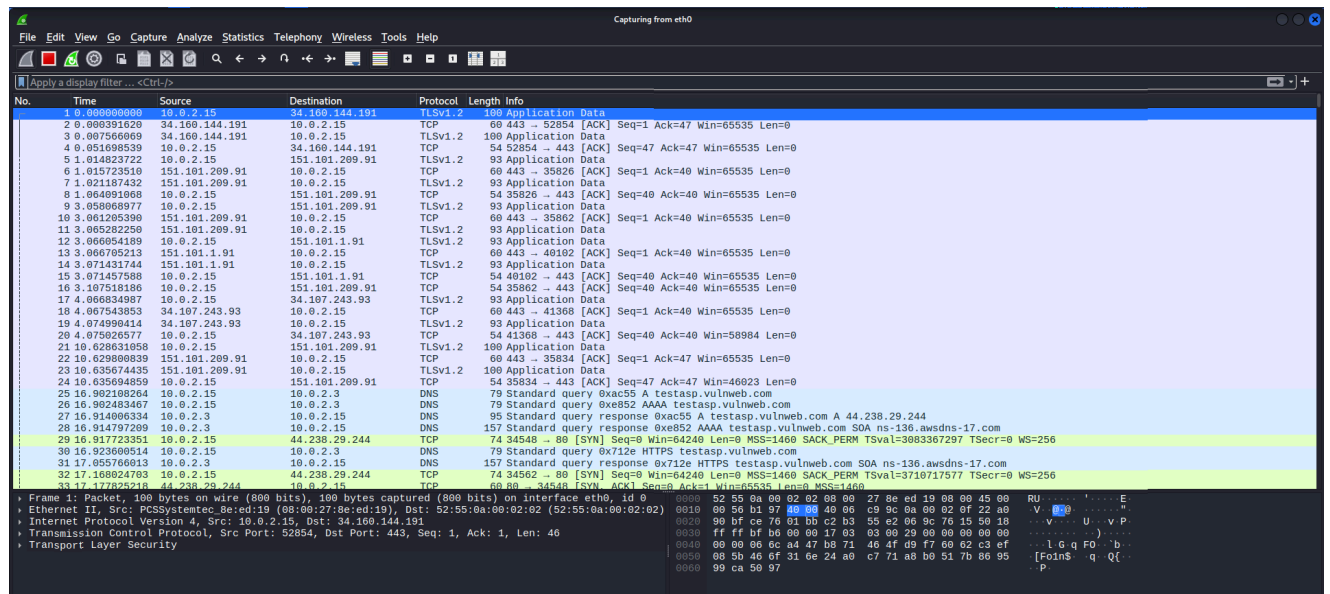


Fig 6: Capturing Network Packets Using Wireshark

Step 4: Capture the Credentials

The login request was opened, and the HTML Form URL Encoded section showed the username and password sent by the website.

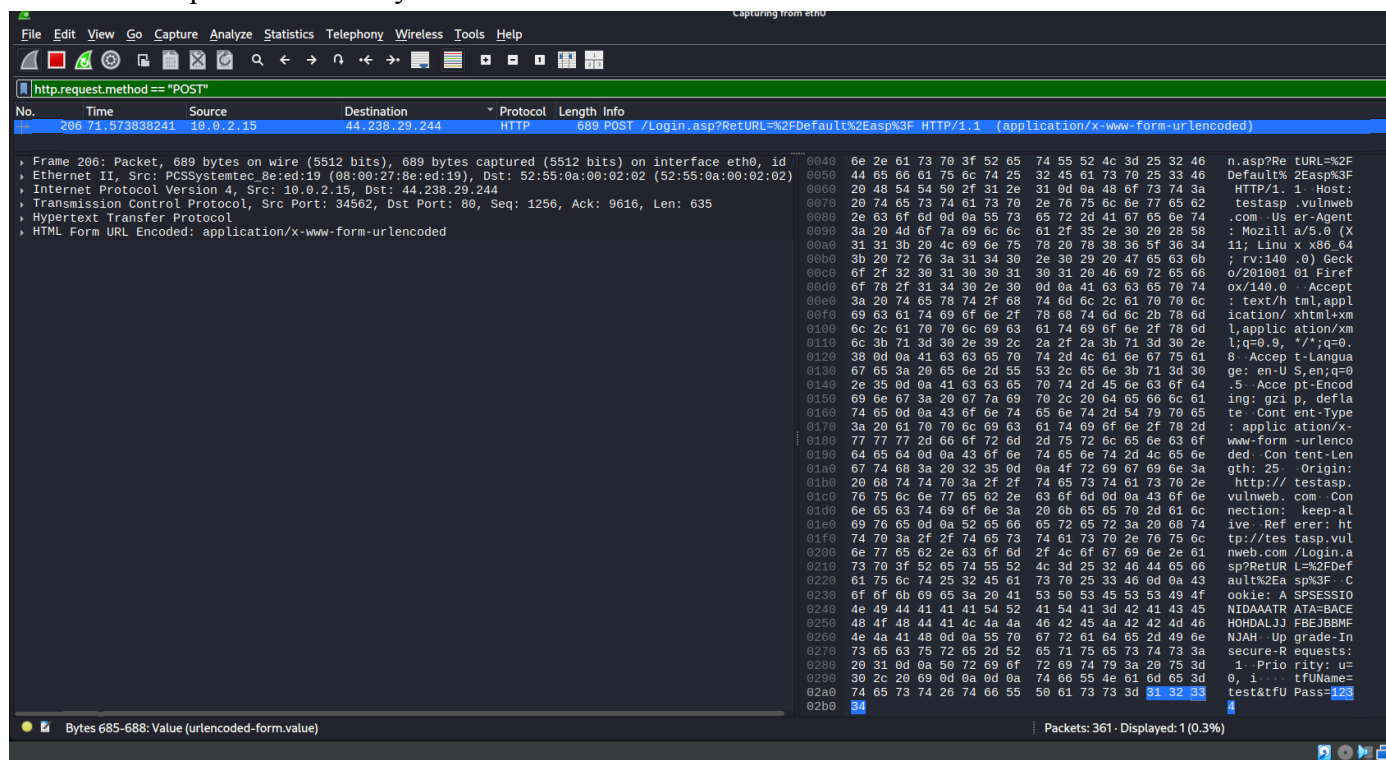


Fig 7: Captured HTTP Login Request

Captured credentials:

Username: test

Password: 1234

Analysis

The network traffic generated during the login process was successfully captured and analyzed using Wireshark. By inspecting the HTTP POST request and examining the HTML Form URL Encoded data, the transmitted username and password were clearly visible in plaintext. This demonstrates that the target website sends authentication data over an unencrypted HTTP connection, allowing anyone monitoring the same network to intercept sensitive credentials.

Impact

1. Login credentials transmitted over HTTP can be intercepted by attackers on the same network.
2. Sensitive user information may be compromised if attackers successfully capture authentication data.
3. Unencrypted communication increases the risk of Man-in-the-Middle (MITM) attacks and credential theft.

Mitigation Steps

1. Use HTTPS instead of HTTP to encrypt all communication between the client and the server.

2. Enable Multi-Factor Authentication (MFA) to provide an additional layer of account security.
3. Regularly monitor network traffic and server logs for suspicious login attempts or unauthorized access.
4. Keep web servers and applications updated with the latest security patches.

Resources Used: Kali Linux, Wireshark, Web Browser (Google Chrome), Target website: <http://testasp.vulnweb.com/>